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Using machine learning in photovoltaics to create smarter and cleaner energy generation systems: A comprehensive review

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ABSTRACT

Photovoltaic (PV) technologies are expected to play an increasingly important role in future energy production. In parallel, machine learning has gained prominence because of a combination of factors such as advances in computational hardware, data collection and storage, and data-driven algorithms. Against this backdrop, we provide a comprehensive review of machine learning techniques applied to PV systems. First, conventional methods for modeling PV systems are introduced from both electrical and thermal perspectives. Then, the application of machine learning to the analysis of PV systems is discussed. We focus on reviewing the use of machine learning algorithms to predict performance and detect faults, and on discussing how machine learning can help humanity to achieve a cleaner environment in the worldwide drive towards carbon neutrality. This review also discusses the challenges to and future directions of using machine learning to analyze PV systems. A key conclusion is that the use of machine learning to analyze PV systems is still in its infancy, with many small-scale PV technologies, such as building integrated photovoltaic thermal systems (BIPV/T), not yet benefiting fully in terms of system efficiency and economic viability. The wider application of machine learning to PV systems could therefore forge a shorter path towards sustainable energy production.

1. Introduction

As the world's population grows, energy production is rising to meet the increasing demand (Abdel-Basset et al., 2021). This leads to serious environmental and economic concerns, motivating governments and policy makers to invest in renewable energy systems, especially those based on solar technology (Darbari and Rashidi, 2022). This has several advantages, including worldwide availability, as well as the ability to be deployed in a large range of installed capacities (Gagliano et al., 2019).

Of the various technologies available to capture solar energy, photovoltaic (PV) systems make a considerable contribution to the overall market (A ğ bulut et al., 2020). There are various plans to increase the installed PV capacity in various parts of the world (Akyol et al., 2021). In a PV system, solar radiation is converted directly into electrical power, while some of the dissipated heat could be recovered and used for a variety of other secondary applications (Dehghan et al., 2021).

Locations at which PV systems can function effectively, especially in urban areas, are limited, so that their selection requires careful consideration (Ayan and Toylan, 2021). In addition, owing to shading, and various other issues, a PV system might operate poorly, requiring appropriate control and fault detection systems (Pan et al., 2020). Conventional technologies face several challenges and there has been remarkable progress made in the field of computer science and statistics. Therefore, the application of machine learning has become widespread in both the design and operation of PV infrastructure over recent years (Ni ž eti ć et al., 2021a).

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Nomenclature

Symbols

A	Area (m^2)
E_g	Energy gap
G	Irradiance ($W.m^{-2}$)
I_o	Diode saturation current (A)
I_{ph}	Photocurrent (A)
N_s	Number of cells connected in series
P	Power (W)
q	Electric charge (C)
R_p	Parallel resistance (Ω)
Rs	Series resistance (Ω)
T	Temperature (K)
V_t	Thermal voltage of diode (V)
V_w	Wind velocity

Greek symbols

β	Temperature coefficient ($\%.^oC^{-1}$)
η	Efficiency (%)
γ	Power coefficient ($\%.^oW^{-1}$)
φ	Relative humidity

Subscripts

amb	Ambient
module	Module
ref	Reference

Abbreviations

ANN	Artificial neural network
BIPVT	Building integrated photovoltaic thermal
BPNNs	Back-propagation neural networks
CNN	Convolutional neural Networks
DT	decision tree
EVA	Ethylene vinyl acetate
FL	Federated learning
GLRT	generalized likelihood ratio test
GNN	Graph neural networks
GPR	Gaussian process regression
IoT	Internet of Things
KNN	K-Nearest Neighbor
LCA	Life cycle assessment
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MC	Mono crystalline
MLP	Multi layer perceptron
MRE	Mean relative error
MR-ESN	Multiple reservoirs echo state network
NMOT	Nominal module operating temperature
NOCT	Nominal cell operating temperature
PC	Poly crystalline
PCA	Principal Component Analysis
PNN	propagation neural network
PV	Photovoltaic
RBF	Radial basis function
RKRF	Reduced-Kernal Random Forest
RMSE	Root mean squared error
STC	Standard test condition
SVM	Support vector machine
TCT	Total cross-tied
UAV	Unmanned aerial vehicles
WT	Wavelet transform

During the design of a PV system, machine learning has been used to identify more accurate ways to optimize the thermal and electrical behavior by addressing phenomena not covered by conventional modeling approaches (Liu et al., 2020). In controlling PV systems, machine learning methods have been used for fault detection, but they could also be used to identify improved tracking strategies (Hadipour et al., 2021). The increasing prominence of machine learning has attracted numerous researchers to this field, whose work has been covered by several recent authoritative reviews.

For example, Akhter et al. (2019) reviewed different methods to predict the performance of a PV module. In that study, various aspects, including the time resolution of the employed data, were considered. In addition, several studies done between 2007 and 2018 with the aim of using machine learning methods, such as artificial neural networks (ANNs) and support vector machines (SVMs), were also covered. Another review, by Berghout et al. (Berghout, Benbouzid, Ma, Djurovi ć, Mouss), focused on the application of machine learning tools to monitor the performance of PV systems. In that review (Berghout, Benbouzid, Ma, Djurovi ć, Mouss), the use of machine learning methods to detect various faults in PV systems – such as bridging, shading, and line-to-line problems – was investigated. Another review paper is that by Kurukuru et al. (Kurukuru, Haque, Khan, Tripathy), who focused on integrating thermal imaging and machine learning techniques. In addition, a review paper by Yahya et al. (2022) explored the subject of PV module fault inspection using thermal-imaging and machine learning. That study focused more on thermal-imaging methods than machine learning.

Table 1 contains an overview of review papers focusing on the application of machine learning to PV systems analysis. Several important research gaps can be identified:

- In most studies, a literature review was performed that considered only one of the many possible applications of machine learning to PV systems, i.e. performance prediction, or fault detection, etc. Thus, there is insufficient focus on different applications, and future perspectives should be formulated by taking various applications into account.
- Another drawback is that conventional modeling approaches have yet to be benchmarked thoroughly. In other words, previous reviews have yet to compare conventional and machine learning approaches systematically.
- The focus of previous reviews has mainly been on technical aspects, such as the algorithm used, the training dataset, the validation dataset, and the type of fault detection. The perspectives formed have thus also focused on the technical aspects, with insufficient focus on environmental and sustainability goals.

Consequently, this provides the motivation for the present review paper:

- The focus is on published research involving both performance prediction and fault detection, rather than on only one of these goals. This should give a broader insight for researchers, policy makers, and investors, facilitating the application of machine learning methods to PV systems and offering valuable commentary on current trends, major barriers, and future perspectives and opportunities.
- In addition to machine learning approaches, conventional methods of analyzing PV systems from both electrical and thermal perspectives are also reviewed. This enables a balanced comparison of the advantages and disadvantages of machine learning approaches in relation to conventional methods.

Table 1
Summary of previous work involving the application of machine learning to PV systems.

Study	Year	A brief description	The points considered in the review	Did the review article consider the application of machine learning in more than one application?	Did the review article consider conventional modeling approaches from both thermal and electrical perspectives?	Did the review article offer insights into the cleaner aspect and sustainable development?
Mellit et al. (Mellit et al., 2018)	2018	Methods to detect faults in various types of PV systems, including individual and hybrid technologies, were reviewed. It covered line-to-line, ground, and arc faults.	The ways for detecting and diagnosis of faults	No	No	No
Das et al. (Das et al., 2018)	2018	According to various classifications, the operation analysis of a variety of PV electricity production methods was done.	Model optimization and PV electricity production	No	No	No
Akhter et al. (Akhter et al., 2019)	2019	Metaheuristic and machine learning approaches were compared. Both critical and systematic reviews were carried out. The results could help to select the best prediction tool.	Application of machine learning methods for estimation of electricity production of PV panels	No	No	No
Dhimish (Dhimish, 2019)	2019	The review discussed and compared 7 innovative maximum power point tracking techniques.	The maximum power point tracking procedures for PV panels on which partial shading and hot spot exist.	No	No	No
Kumaradurai et al. (Kumaradurai, Teekaraman, Coosemans, Messagie)	2019	The review provided a discussion of different faults in PV systems and ways to detect them. It included different issues such as prediction of healthy condition for a PV panel, DC and AC fault classification, and cost-accuracy-reliability issues.	Application of machine learning approach to detect faults in PV technologies	No	No	No
Mosavi et al. (Mosavi et al., 2019)	2019	The study covered the application of 10 machine learning methods, including artificial neural network (ANN), support vector machine (SVM) to predict the performance of various energy systems, including PV panels.	Utilization of machine learning modeling approach for energy systems	No	No	No
de Freitas Viscondi et al. (de Freitas Viscondi and Alves-Souza, 2019)	2019	Around 100 papers were reviewed, while a complete analysis was carried out for 38 investigations. The review searched for the most suitable ways to use big data for PV electricity production estimation.	Big data to estimate PV electricity production	No	No	No
Houssein et al. (Houssein, 2019)	2019	The review study discussed the application of different machine learning and metaheuristic approaches to predict the performance of several types of renewable energy systems, including solar, biogas, biomass, and wind energies.	Utilization of metaheuristic and machine learning approaches for renewable energy systems	No	No	No
Mosavi and Bahmani (Mosavi and Bahmani, 2019)	2019	The investigation covered a variety of subjects, including types of estimation, utilization coverage, and modeling techniques. An upward trend in the application and estimation accuracy of machine learning techniques was found.	Application of machine learning techniques for forecasting of consumption of different types of energy, including renewables	No	No	No
Darío Obando et al. (Obando et al., 2019)	2019	Different ways to predict the amount of received irradiance were reviewed. Machine learning, as well as analytical and numerical techniques were investigated.	Estimation of the amount of the received irradiance by machine learning approaches	No	No	No
Fan et al. (Fan et al., 2019)	2019	From the modeling approaches considered, twelve were chosen. Then, these were evaluated for	Machine learning and empirical tools for estimation of the irradiance	No	No	No

(*continued on next page*)

Table 1 (*continued*)

Study	Year	A brief description	The points considered in the review	Did the review article consider the application of machine learning in more than one application?	Did the review article consider conventional modeling approaches from both thermal and electrical perspectives?	Did the review article offer insights into the cleaner aspect and sustainable development?
		diverse climatic conditions in China as the case-study.	on the day from the times appears in the sky.			
Abdulmawjood et al. (AbdulMawjood, Refaat, Morsi)	2020	The DC faults in a PV array were introduced, their origins were discussed, and their impact on the PV performance were described. Moreover, explanations of the classification and detection techniques were given.	PV array fault detecting and prediction	No	No	No
Ahmad and Chen (Ahmad and Chen, 2020)	2020	This study listed different machine learning prediction techniques, including supervised ones, and compared them. The goal was to introduce suitable machine learning tools for a variety of applications.	The increasing employing machine learning tools, as well as their utilization in the real time usage	No	No	No
Berghout et al. (Berghout, 2021)	2021	The review provided a framework for discussing different artificial intelligence tools, with various learning methods, including ensemble, reinforcement, transfer, and deep learnings for performance monitoring PV systems.	Monitoring the condition of PV systems	No	No	No
Kurukuru et al. (Kurukuru, Haque, Khan, Tripathy)	2021	Classification of faults in a PV technology using machine learning and thermography tools was discussed	Application of machine learning and thermography tools to classify faults in PV technologies	No	No	No
Hwang et al. (Hwang et al., 2021a)	2021	The work considered a variety of machine learning techniques. Thermography, CNN, and XGBoost were among the methods investigated.	Using machine learning tools to find defects in a PV system	No	No	No
Li et al. (Li et al., 2021a)	2021	Both ANN and hybrid approaches were discussed. The discussion covered several aspects, such as the amount of data and the faults that could be detected.	ANN use for diagnosis and detecting defects in a PV system	No	No	No
Tina et al. (Tina et al., 2021)	2021	Comparison of different perspectives on the performance of a PV system, including economic factors, electricity production, and solar radiation.	Use of machine learning and artificial intelligence approaches to estimate several indicators related to the performance of PV systems	Yes	No	No
Zhou et al. (Zhou et al., 2021)	2021	The review explained different important aspects of machine learning tools. It included feature selection, input parameters, and developing models.	Global received irradiance	No	No	No
Ağbulut et al. (A ğ bulut et al., 2021)	2021	4 machine learning approaches, KNN, ANN, SVM, and deep learning were compared.	Global received irradiance	No	No	No
Bamisile et al. (Bamisile et al., 2021)	2021	Machine learning and deep learning prediction ways were compared. Of the former, 3 techniques, i.e., recurrent neural network (RNN), ANN, and CNN were considered. Support vector and polynomial regressions were also the tools considered in the latter.	Diffuse and global received irradiance	No	No	No
Al-Hajj et al. (Al-Hajj et al., 2021)	2021	Several stacking-based ensembles structures were compared. The tools investigated were KNN, ANN, and support vector regression approaches.	Global received irradiance	No	No	No
Yahya et al. (Yahya et al., 2022)	2022	The review work was divided into three parts. The acquisition of pictures and key fault patterns for various spectra was the first part. The second part included pre-	Inspecting PV systems	No	No	No

(*continued on next page*)

Table 1 (*continued*)

Study	Year	A brief description	The points considered in the review	Did the review article consider the application of machine learning in more than one application?	Did the review article consider conventional modeling approaches from both thermal and electrical perspectives?	Did the review article offer insights into the cleaner aspect and sustainable development?
		processing techniques, while machine learning and deep learning tools, as well as digital image processing tools was the subject of the third part.				
The current work	2022	A broad insight into using both conventional and machine learning approaches to predict the performance of PV systems is provided. Moreover, we review machine learning algorithms for fault detection and diagnosis of PV technologies. Additionally, we discuss how machine learning tools can be used to solve problems, reconfigure, and manage PV systems. In addition to the technical aspects, we provide insights into clean energy and sustainability objectives.	Use of conventional methods, as well as machine learning tools for both performance prediction and fault detection of PV technologies	Yes	Yes	Yes

- Besides the technical aspects of machine learning, this review offers insights into the clean energy and sustainability objectives. In this context, a map for reaching certain environmental targets is discussed.

2. Research methodology

To identify suitable articles for the review, a search was initiated on "Scopus" (Scopus database, 2022) and "Google Scholar" (Google Scholar database) databases using a variety of keywords, including different combinations such as "machine learning", "artificial intelligence", "performance prediction", "modeling", "fault", "defect", "failure", "diagnosis", "classification", "neural network", as well as different machine learning methods (e.g., "artificial neural network (ANN)" or support vector machine (SVM)) and various types of faults (e.g., "short circuit", "degradation", "line to line" kind), in addition to their standard abbreviations.

The most recent article was one with an original submission date of March 1, 2022, while the corresponding revision date was April 10, 2022. Based on the search results, over 150 articles were identified and included in the present review.

3. Conventional methods for performance prediction

PV technologies have been used in a variety of applications. They have been used as stand-alone systems in the form of PV farms. In addition, they have the potential to be integrated with buildings. PV panels can be installed on the wall, on the roof, as the windows, etc. Additionally, some of the heat from a PV panel can be absorbed by a fluid, e.g., air or water, and the heat can be recovered. Such systems are called photovoltaic thermal (PV/T) systems. The methods used to estimate the performance of PV systems can be divided into two groups. The first is the conventional method, which refers to the application of either simple correlations or governing equations to analyze the system performance. The second is machine learning methods. Below, we discuss conventional methods to determine the electrical and thermal parameters of a PV module.

3.1. Thermal modeling

The goal of thermal modeling is to determine the working temperature of a solar module or other related performance metrics.

3.1.1. Nominal operating cell temperature (NOCT)

The NOCT approach is the simplest way to predict the working temperature of a PV module. It needs only the ambient temperature (T_{amb}) and the received solar radiation (G). In the NOCT approach, T_{module} is predicted using Eq. (1) (Standard IJDq and approval t. 61215, 1215):

$$T_{module,NOCT} = T_{amb} + \frac{G}{G_{ref}}\left(T_{NOCT} - T_{ref}\right) \quad (1)$$

The NOCT temperature (T_{NOCT}) is a parameter available for each module. The subscript 'ref' denotes the reference condition. At the reference condition, the temperature and irradiance values are 20 °C and 800 W m^{-2}, respectively. The reference condition is not the same as the standard test condition (STC), at which the temperature and irradiance are 25 °C and 1000 W m^{-2}, respectively.

3.1.2. Correlation techniques

Eq. (1) could easily be applied to each solar module. However, it has some shortcomings, including:

- Modules from the same product family usually have identical values of T_{NOCT}. Therefore, the NOCT approach predicts the same value for all dimensions of a solar module product family. Owing to variations in dimension and hence the heat transfer rate, T_{module} is not the same across different module sizes within a product family, even when T_{amb} and G remain fixed.
- In addition to T_{amb} and G, the wind velocity (V_w) and relative humidity (φ) can also affect T_{module}. However, the NOCT approach does not take these into account.

In recent years, other correlations have also been proposed, based on experiments. Table 2 lists some of the most important ones. As can be seen, various functions have been developed. Of these the nominal module operating temperature (NMOT) is one of the most widely adopted, and was assigned an IEC number of 61853–2 for PV rating.

Our search found that few of the many correlations proposed account for the effect of relative humidity. However, in some recent studies, e.g. Sohani et al. (Sohani and Sayyaadi, 2020), the relative humidity was considered. As good practice, the experimental data used to determine

Table 2
Key correlations used to predict the temperature of a PV module (Sohani and Sayyaadi, 2020).

Study	Year	Ambient temperature (T_{amb})	Solar radiation (G)	Wind speed (V_w)	Relative humidity (φ)	Proposed function for working temperature prediction (T_{panel})
Ross (Ross, 1976)	1976	Yes	Yes	No	No	$T_{panel} = T_{amb} + 0.024G$
Risser and Fuentes (Risser and Fuentes, 1984)	1984	Yes	Yes	Yes	No	$T_{panel} = 0.899T_{amb} + 0.026175G - 0.00000754G^2 - 1.30V_w + 3.8621$
Scott (Schott)	1985	Yes	Yes	No	No	$T_{panel} = T_{amb} - 1 + 0.028G$
Servant (SERVANT, 1986)	1986	Yes	Yes	Yes	No	$T_{panel} = T_{amb} + 0.016G(1 + 0.030T_{amb})(1 - 0.085V_w)$
Chenni et al. (Chenni et al., 2007)	2007	Yes	Yes	Yes	No	$T_{panel} = T_{amb} + 0.0138G(1 + 0.031T_{amb})(1 - 0.042V_w)$
Mondol et al. (Mondol et al., 2007)	2007	Yes	Yes	No	No	$T_{panel} = T_{amb} + 0.031G - 0.058$
Skoplaki et al. (Skoplaki et al., 2008)	2008	Yes	Yes	Yes	No	$T_{panel} = T_{amb} + \left(\frac{0.25}{5.7 + 3.8V_w}\right)G$
Skoplaki et al. (Skoplaki and Palyvos, 2009)	2009	Yes	Yes	Yes	No	$T_{panel} = T_{amb} + \left(\frac{0.32}{8.91 + 2V_w}\right)G$
King (King)	2009	Yes	Yes	Yes	No	$T_{panel} = T_{amb} + \frac{G}{G_{ref}}(0.0712V_w^2 - 2.411V_w + 32.96)$
Akyuz et al. (Akyuz et al., 2012)	2012	Yes	Yes	Yes	No	$T_{panel} = 0.95T_{amb} + 0.025G - 0.3V_w + 3.1$
Almaktar et al. (Almaktar et al., 2013)	2013	Yes	No	No	No	$T_{panel} = 1.411T_{amb} - 6.414$
Kalogirou (Kalogirou, 2013)	2013	Yes	Yes	No	No	$T_{panel} = 1.14(T_{amb} - T_{ref}) + 0.0175(G - 300) + 30$
Muzathik et al. (Muzathik, 2014)	2014	Yes	Yes	Yes	No	$T_{panel} = 0.943T_{amb} + 0.0195G - 1.528V_w + 0.3529$
Mora Segado et al. (Mora Segado et al., 2015)	2015	Yes	Yes	Yes	No	$T_{panel} = T_{amb} + 0.79\left[\frac{G}{G_{ref}}(T_{panel,NOCT} - 20)\right] - 1.52(V - V_{ref})$
Standard 61215 (Standard IJDq and approval t. 61215, 1215)	2016	Yes	Yes	No	No	$T_{panel,NOCT} = T_{amb} + \frac{G}{G_{ref}}(T_{NOCT} - T_{ref})$
Coskun et al. (Coskun et al., 2016)	2016	Yes	Yes	Yes	No	$T_{panel} = 1.4T_{amb} + 0.01(G - 500) - V_w^{0.8}$
Tselepis and Tripanagnostopoulos (Tselepis, Tripanagnostopoulos)	2016	Yes	Yes	No	No	$T_{panel} = 30 + 0.0175(G - 150) + 1.14(T_{amb} - 25)$
PV rating IEC 61853-2 (CODE, PRIX)	2016	Yes	Yes	Yes	No	$T_{panel,NMOT} = T_{amb} + G\frac{1}{u_0 + u_1V_w}$
Lasnier (Lasnier, 2017)	2017	Yes	Yes	No	No	$T_{panel} = 30.006 + 0.0175(G - 300) + 1.14(T_{amb} - 25)$
Coskun et al. (Coskun et al., 2017)	2017	Yes	Yes	Yes	No	$T_{panel} = T_{amb} + 0.01632578G - 0.8V_w + 2.273$
Kurtz et al. (Kurtz et al., 2399)	2019	Yes	Yes	Yes	No	$T_{panel} = T_{amb} + G\ e^{(-3.473 - 0.0594V_w)}$
Sohani et al. (Sohani and Sayyaadi, 2020)	2020	Yes	Yes	Yes	Yes	$T_{panel} = \frac{1}{3}\left\{aT_{amb} + b\left(\frac{G}{G_{ref}}\right)^c + d\varphi^e + fV_w + g\right\} - 273.15$

the coefficients of a correlation should cover at least six months of a year, as mentioned in several references such as (S á nchez-Palencia et al., 2019). The need for long-term experimental data, which is costly in terms of both time and finance, could be the most significant drawback to the above correlation approaches.

3.1.3. One dimensional (1D) numerical approach

In the 1D numerical approach, a PV module is considered as a number of layers (Fig. 1). Then, for all the points on a layer, the same temperature is assumed. Solving the governing equations gives the temperature of each layer.

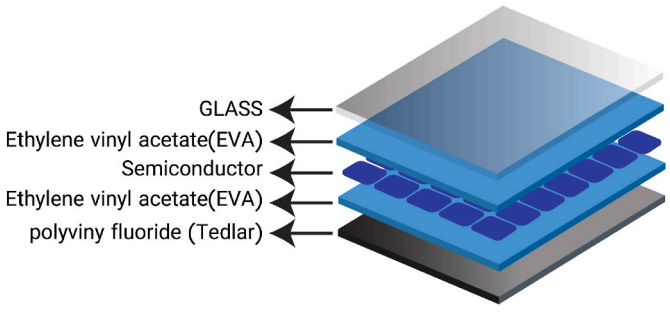


Fig. 1. The five layers of a PV module considered in 1D numerical modeling (Sohani et al., 2021a).

Compared with NOCT, NMOT and other correlation-based approaches, 1D numerical modeling is more accurate because it uses the governing equations. However, the number of inputs required – e.g. the thickness of the layers, their isobaric heat capacities and densities – is greater than in the previously mentioned approaches. An overview of the studies in which 1D numerical modeling has been either developed or utilized is provided in Table 3. This table has been compiled using the information available in the literature review of Ref. (Sohani et al., 2021a). It can be seen that the temperature dependency of the layers' properties, including the isobaric heat capacity and density, have been neglected in most studies. Nonetheless, as shown in (Sohani et al., 2021a), accounting for the temperature dependency of the indicated properties can enhance the modeling accuracy. It is worth noting that the number of studies in which 1D numerical model has been used is less than that for other approaches.

In addition to requiring a higher number of input parameters, a longer run-time and the need to have access to the governing equations are generally taken to be the most serious challenges to the 1D numerical approach. A particular drawback is that 1D modeling cannot capture the temperature distribution on the module surface, necessitating numerical models with higher dimensions.

3.1.3.1. 2 and 3 dimensional (2D and 3D) numerical techniques. When a 2D or 3D numerical scheme is used, the temperature distribution on the module's surface can be determined by solving the governing equations, e.g. the energy and momentum equations. A list of recent studies

Table 3
An overview of previous work involving 1D numerical modeling of a PV system.

Authors	Year	System	Analysis approach	Was the temperature dependency of thermal properties considered?
Tomar et al. (Tomar et al., 2018)	2018	Building integrated photovoltaic thermal (BIPVT) units	Parametric study	No
Singh et al. (Singh et al., 2019)	2019	Photovoltaic thermal technology	Parametric study	No
Gu et al. (Gu et al., 2019)	2019	Photovoltaic technology	Parametric study	No
Osma-Pinto and Ordonez-Plata (Osma-Pinto and Ord ón ez-Plata, 2020)	2020	Photovoltaic thermal technology	Parametric study	No
Shahsavar et al. (Shahsavar et al., 2020)	2020	Photovoltaic thermal technology	Parametric study	No
Shahverdian et al. (Shahverdian et al., 2021a)	2021	Photovoltaic thermal technology	Optimization	No
Sohani et al. (Sohani et al., 2021a)	2021	Photovoltaic technology	Parametric study	Yes; all layers except for Tedlar layer

involving 2D or 3D numerical modeling of a PV system is presented in Table 4. Table 4 takes advantage of the information provided in the literature review in (Sohani et al., 2021b).

As the 2D and 3D numerical approaches solve the governing equations to provide the temperature distribution on the module surface, they can be considered more accurate than the 1D numerical, NOCT and other correlation approaches. However, they require more computational cost and time, as well as knowledge of a number of different processes, such as heat and mass transfer. They also need more input parameters than others.

3.2. Electrical modeling

The purpose of electrical modeling is to determine the power as the main output of the system. Voltage and current are also parameters which can be determined by electrical modeling.

3.2.1. Correlation approach

In most studies, when the goal is to extract power, Eq. (2) is first used to determine the efficiency. Then, to calculate the power, the efficiency (η) is multiplied by the received solar radiation (G) and module area (A_{module}), which are known parameters (Rashidi et al., 2021):

$$P = \eta G A_{module} \tag{2}$$

$$\eta = \eta_{ref}\left\{1 - \beta_{ref}\left(T_{module} - T_{ref}\right) + \gamma \log_{10}(G)\right\} \tag{3}$$

T_{module} is a known parameter, which can be found using thermal modeling. In addition, η_{ref} is the module efficiency in the reference condition, while γ is the coefficient for the condition with maximum power. Both η_{ref} and γ are usually available from the manufacturers.

3.2.2. Equivalent circuit method

If the focus is on electrical parameters other than power, the equivalent circuit method can be deployed. In this method, the electrical performance of the module is modeled using an equivalent circuit containing several diodes and resistors. Three important items in each equivalent circuit are:

- Photocurrent (I_{ph}): This quantifies the current generated by the received solar radiation.
- Series resistance (R_s): Some of the energy generated by the solar module is dissipated by passing current through the semiconductor and metallic connections.

Table 4
List of recent studies involving 2D or 3D numerical modeling of a PV system.

Study	Year	Module type	Modeling dimension	The compared engineering methods	Temperature distribution related criteria
Zondag et al. (Zondag et al., 2002)	2002	Poly crystalline (PC)	1D, 2D, and 3D	N.A.	N.A.
Usama Siddiqui et al. (Usama Siddiqui et al., 2012)	2012	MC	3D	NOCT	Module mean temperature
Pierrick et al. (Pierrick et al., 2015)	2015	Mono crystalline (MC)	3D	N.A.	Module mean temperature
Guarracino et al. (Guarracino et al., 2016)	2016	MC	3D	N.A.	N.A.
Paradis et al. (Paradis et al., 2017)	2017	MC	2D	N.A.	Module mean temperature
Hosseinzadeh et al. (Hosseinzadeh et al., 2018)	2018	MC	3D	N.A.	N.A.
Nasrin et al. (Nasrin et al., 2018)	2018	PC	3D	N.A.	Module mean temperature
Amanlou et al. (Amanlou et al., 2018)	2018	MC	3D	N.A.	Module mean temperature
Yamamoto et al. (Yamamoto, Wagi, Nanno)	2018	MC	2D	N.A.	N.A.
Fayaz et al. (Fayaz et al., 2019)	2019	PC	3D	N.A.	Module mean temperature
Maadi et al. (Maadi et al., 2019)	2019	PC	3D	N.A.	Module mean temperature
Kazemian et al. (Kazemian et al., 2019)	2019	MC	3D	N.A.	Module mean temperature
Atsu and Dhaundiyal (Atsu and Dhaundiyal, 2019)	2019	Poly and mono crystalline, as well as amorphous thin film Si-based modules,	2D	NOCT	N.A.
Hu et al. (Hu et al., 2020)	2020	MC	2D	N.A.	N.A.
Salari et al. (Salari et al., 2020)	2020	MC	3D	N.A.	Module mean temperature
Abd El-Samie et al. (El-Samie et al., 2020)	2020	PC	3D	N.A.	Module mean temperature
Salameh et al. (Salameh et al., 2021)	2021	PC	3D	NOCT	Module mean temperature
Sohani et al. (Sohani et al., 2021b)	2021	MC and PC	1D, 2D, and 3D	NOCT and NMOT	Changes in prediction error by variation of σ

- Parallel resistance (R_P): Parallel resistance, also known as shunt resistance, is used to capture several phenomena, such as passing current from the module's edges, crystal geometry holes and non-idealities.

3.2.2.1. Single diode model. Also called the one diode approach, this method considers an equivalent circuit like that in Figure (2a). This equivalent circuit is composed of a series resistance, a shunt resistance, and a current source that quantifies the generated photocurrent. In addition to these items, there is also a diode in the circuit. Some of the current generated by sunlight goes into the diode, which decreases the received voltage at the terminals. According to electrical governing rules, the relationship between the current and voltage for this single diode model is (Ni ž eti ć et al., 2021b):

$$I = I_{ph} - I_0\left[\exp\left(\frac{1}{V_t}\left(\frac{V}{N_s} + I.R_s\right)\right) - 1\right] - \frac{1}{R_p}\left(\frac{V}{N_s} + R_s I\right) \tag{4}$$

Besides I_{ph}, R_s, and R_P, there are three other important parameters in the equations. They are (Ni ž eti ć et al., 2021a):

Thermal voltage of diode (V_t): This describes the thermal driving force of electrons in a semi-conductor. However, this movement is in the opposite direction to the photocurrent.

Diode saturation current (I_0): This parameter describes the movement of minority of charge carriers in a semi-conductor from neutral to the depletion area.

Number of connected cells in series in a module (N_S): A module is built by connecting a number of cells that are electrically in series.

In addition to the voltage and current, the other parameters in Eq. (4) are obtained from Eqs. (5)–(9):

$$R_p = \frac{G_{STC}}{G} R_{p,STC} \tag{5}$$

$$R_s = R_{s,STC} \tag{6}$$

$$I_0 = I_{0,STC}\left(\frac{T_{module}}{T_{module,STC}}\right)^3 \exp\left[\frac{qE_g}{K}\left(\frac{1}{T_{module,STC}} - \frac{1}{T_{module}}\right)\right] \tag{7}$$

$$V_t = \frac{T_{module}}{T_{module,STC}} V_{t,STC} \tag{8}$$

$$I_{ph} = \frac{G}{G_{STC}}\left\{I_{ph,STC} + \alpha\left(T_{module} - T_{module,STC}\right)\right\} \tag{9}$$

To determine the values in the condition investigated, values at STC are required. The values of I_0, V_t, and I_{ph} at STC are calculated from Eqs. (10)–(12) (Raju et al., 2022):

$$I_{0,STC} = I_{sc,STC}\exp\left(\frac{-V_{oc,STC}}{N_s V_{t,STC}}\right) \tag{10}$$

$$V_{t,STC} = \frac{\beta\ T_{module,STC} - V_{oc,STC}}{\frac{N_s T_{module,STC}\ \alpha}{I_{ph,STC}} - 3N_s - \frac{E_g N_s}{K T_{module,STC}}} \tag{11}$$

$$I_{ph,STC} = I_{sc,STC} \tag{12}$$

The parameters on the right-hand side of Eqs. (10)–(12) are known. They are either the module characteristics available from the module catalogue (such as $I_{sc,STC}$, $V_{oc,STC}$, β, and α) or constant parameters (such as E_g and K). In addition, the value of the series resistance at STC is computed by solving Eq. (13), in which this parameter is the only unknown variable:

$$I_{mp,STC} = I_{ph,STC} - I_{0,STC}\left[\exp\left(\frac{V_{mp,STC} + I_{mp,STC}R_{s,STC}}{N_s V_t}\right) - 1\right] - \frac{\left(V_{mp,STC} + I_{mp,STC}R_{s,STC}\right)\left[-N_s V_{t,STC} I_{mp,STC} + \left(V_{mp,STC} - I_{mp,STC}R_{s,STC}\right)\left(I_{sc,STC} - I_{mp,STC}\right)\right]}{\left(-I_{mp,STC}R_{s,STC} + V_{mp,STC}\right)\left(V_{mp,STC} - N_s V_{t,STC}\right)} \tag{13}$$

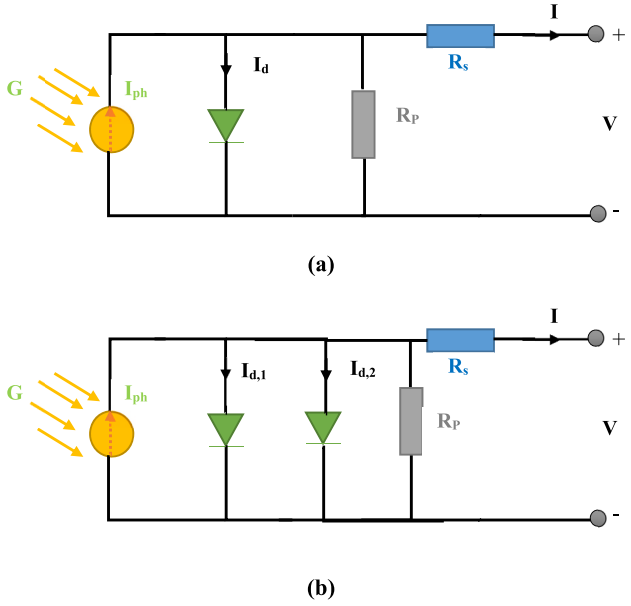


Fig. 2. The diode equivalent circuit; (a) single diode approach; (b) double diode approach.

The shunt resistance at STC can also be determined from Eq. (14):

$$R_{p,STC} = \frac{\left(-I_{mp,STC}R_{s,STC} + V_{mp,STC}\right)\left(V_{mp,STC} - N_s V_{t,STC}\right)}{-N_s V_{t,STC} I_{mp,STC} + \left(V_{mp,STC} - I_{mp,STC}R_{s,STC}\right)\left(-I_{mp,STC} + I_{sc,STC}\right)} \tag{14}$$

3.2.2.2. Double diode model. An important metric for a PV module, which is like a diode, is the ideality factor. On the one hand, at a lower voltage range the ideality factor is close to 2 because junction recombination is dominant. On the other hand, at a higher voltage range the ideality factor approaches unity because the dominant process is recombination by the bulk area and surfaces of the PV module. In the single diode model, the ideality factor is considered constant. Nonetheless, by adding another diode in parallel with the one in the single diode model, improved accuracy is achieved (Fig. 2)). The equations for the double diode model are like those for the single diode model, with the only modifications arising from adding the second diode. They can be found in various references, such as Babu and Gurjar (2014), Shannan et al. (Shannan, Yahaya, Singh), Chennoufi et al. (2021), and Sangeetha et al. (2017).

3.3. Coupling thermal and electrical models

As the governing equations shows, on the one hand, to determine the temperature of the module, the electrical power should be known. On the other hand, obtaining the electricity produced requires the module working temperature values. Therefore, the thermal and electrical models must be solved together, i.e., they should be coupled.

3.4. Problems of the conventional approaches

Despite being used widely, the conventional approaches suffer from some drawbacks. The first is that the simple correlations for thermal modeling do not take into account a number of effective parameters such as relative humidity, while they are mostly accurate for one module, and they cannot be generalized (Shahverdian et al., 2021b). In addition, the more advanced thermal modeling approaches, such as 1D, 2D, and 3D numerical approaches, do not consider some issues like hot spots on the surface of the module, which leads to a high difference between the actual and predicted working temperature values. Moreover, there have been several challenges to parameter estimation by the equivalent circuit method, presenting its use with serious difficulties. As a result, machine learning approaches are being increasingly used for PV systems. Not only are they employed to predict performance from both the electrical and thermal aspects, they are also used for fault detection, as will be discussed below.

4. Using machine learning approach for PV systems

As has just been mentioned, machine learning approaches are increasingly being used for PV systems, for two reasons. One is performance prediction, and the other is fault detection. Reviewing the literature shows that in most studies outdoor tests have been used.

4.1. Performance prediction

Performance prediction is one of the most important applications of machine learning approaches for PV installations. This can be done with the aim of determining one or more parameters from the following list (Fig. 3):

- Working temperature,
- Voltage,
- Current,
- Power.

The input parameters of those models have usually been meteorological parameters, as well as a number of module specifications (Jur č evi ć et al., 2021). As an example of the investigations conducted, Han et al. (2019) considered a ELM-based multi system to predict the power generated at a photovoltaic plant in Beijing, China. Using the prediction tool they developed, the mean absolute errors were 1.70% during the hot season and 2.13% in spring, but using the single model, the corresponding values were 1.81% and 2.43%, respectively. Therefore, the prediction tool they developed estimated the power more accurately.

In (Yao et al., 2019), a model that had been developed using a multiple-reservoir echo-state network was used to predict photovoltaic power. In the model, referred to as MR-ESN, the reservoir parameters were optimized via the quasi-Newtonian algorithm. According to the results, the MAPE value was 0.00195%, which is very close to zero. This model was more accurate than others such as conventional neural networks, wavelet transforms (WT), support vector machines (SVM), and support vector regression (SVR-ANN).

Zazoum et al. (Zazoum, 2022) applied Gaussian process regression (GPR) and SVM approaches to predict the power of a PV system, and compared them. MAE, R^2, and RMSE were the error-related criteria chosen. The results revealed that between the two approaches used, GPR showed better performance, with values of MAE, R^2, and RMSE of 5.302, 0.98, and 7.967, respectively.

Al-Dahidi et al. (2018) used an ELM algorithm in a MLP network to predict the power of a PV plant in Amman, Jordan. The lowest mean absolute error was 1.08% in June, and the highest mean absolute error was 18.83% in February and March. SVM with big data analysis was

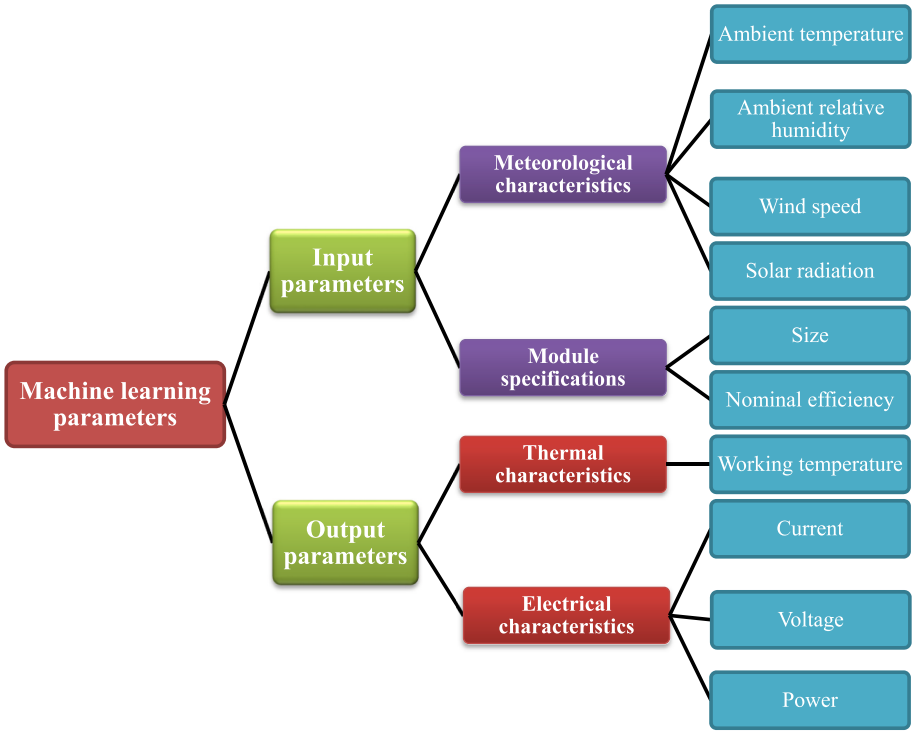


Fig. 3. Showing the considered input and output parameters of machine learning methods for performance prediction of a PV power production system.

employed by Preda et al. (2018) to determine the power of a PV module.

To predict the performance of a PV system located in Beijing, both SVM and MLP methods were used in Ref. (Liu et al., 2018). The model used was very successful, especially for different circumstances with mist and fog. Air temperature and relative humidity were among the inputs to the model. A spatiotemporal graph neural network was used by Karimi et al. (Karimi et al., 1532). The goal of the network was to forecast the performance of the PV system 15–120 min ahead. In that investigation, MAE was one of the parameters used to evaluate the error.

A new method based on satellite and NWP data was developed to forecast the electricity generated by a PV system in Italy (Pierro et al., 2017). The method was verified at a small-scale PV solar power plant in Tyrol, Italy. Over a 4-h time horizon, the RMSE was between 5% and 7% and for one day, it was in the range of [7%–7.5%]. Leva et al. (2017) provided an MLP-based model to predict PV performance using weather data. This method worked better on sunny days than on cloudy days. Das et al. (2017) used a SVM-based model to predict PV power generation. In Malaysia, this model was validated in different weather conditions including clear, cloudy and rainy days. The mean absolute error was 34.57%.

Paulescu et al. (2017) predicted the electricity produced by two PV systems in Catania and Milan in Italy using a fuzzy-logic approach. The average absolute errors were 0.64 and 0.56 kW for Milan and Catania, respectively. In that study, the model being studied had better accuracy in the hot season than the cold. Benhmed et al. (Benhmed et al., et al.) estimated the power value of a PV module under Qatari climatic conditions. In that study, machine learning was utilized. The approaches employed incorporated linear regression and M5P tree models. Mean and absolute, as well as their square roots were used as the error-related parameters. Khandakar et al. (2019) used ANN in addition to the M5P decision tree, and Gaussian processes and simple linear regression techniques to predict the power of a PV unit. The results demonstrated that ANN had a much lower error, with a root mean square error of 2.1436.

A fuzzy T-S method was presented in (Liu et al., 2017) to predict PV performance. The inputs to this method were meteorological parameters. The results from the prediction tool were compared with other methods. The results showed that the resulting prediction tool had the lowest absolute error in summer and the highest average error in spring, 9.77% and 30%, respectively.

Li et al. (2016) used a hierarchical approach to determine system performance. Various parameters such as power and system geometry were used as inputs. Methods based on ANN and SVR were found to be more accurate than others. Moreover, Sohani et al. (2020) employed ANN to predict the electrical characteristics of two PV modules. One poly- and one mono-crystalline module, each of 50 W capacity, were studied, with almost identical dimensions. Relative humidity was one of the inputs for the ANN. A parametric study was then conducted using the ANN to discover the impact of relative humidity alone on the system operation.

In addition to the temperature, current, voltage, and power of module, machine learning tools have been also employed to predict the thermoelectrical characteristics of a module. In the study by Sohani et al. (2022) the authors developed ANN to obtain the photocurrent and thermal voltage of the diode. The inputs for the ANN they developed were the same as in their previous studies, i.e., ambient temperature and relative humidity, as well as wind velocity and wind direction. The investigation showed that the mean absolute error values for prediction of photocurrent by the ANN developed were 4.8, 5.7, 5.3, and 5.1% for Tehran, Kuala Lumpur, Ankara, and Rome, respectively. The corresponding values for these cities, which represented dry, tropical, continental, and temperate climates respectively, were 2.4, 3.6, 3.3, and 2.9%, respectively. Consequently, the ANN they developed could be used for different cities worldwide.

4.2. Fault detection, diagnosis and solving that

Several faults can occur in a PV system, as Figure (4) shows. According to several references such as Sabbaghpur et al. (Sabbaghpur Arani and Hejazi, 2016) and Berghout et al. (2021), of the various possible faults in a PV system, six are dominant (Fig. 5):

- Shading,
- Degradation,
- Bypass diode,
- Line to line,
- Open circuit,
- Bridging

In this part, a brief explanation of each item is given, and then machine learning based solutions that have been provided in the literature are introduced. In addition to the above major faults in a PV system,

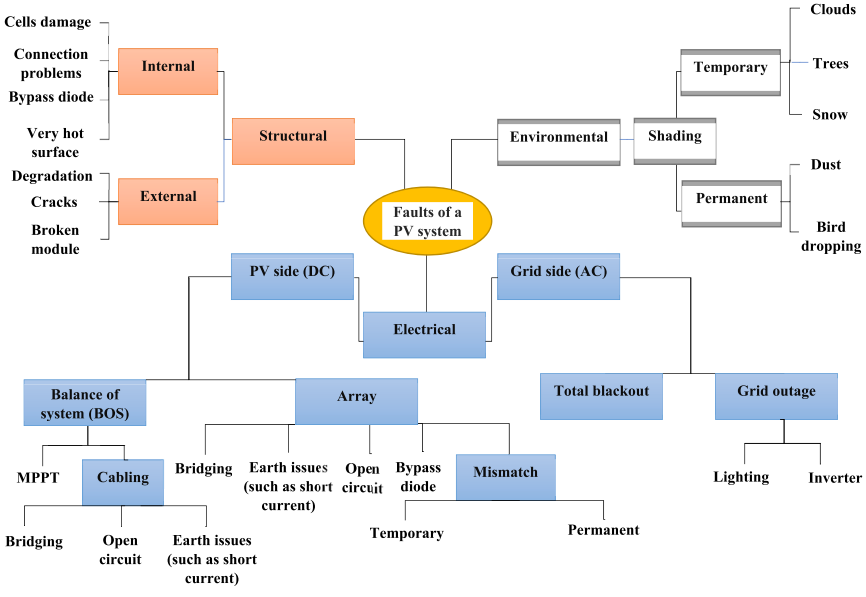


Fig. 4. Classifications of faults in a PV system. The figure has been re-drawn from the figures in (Pillai and Rajasekar, 2018) and (Livera et al., 2019).

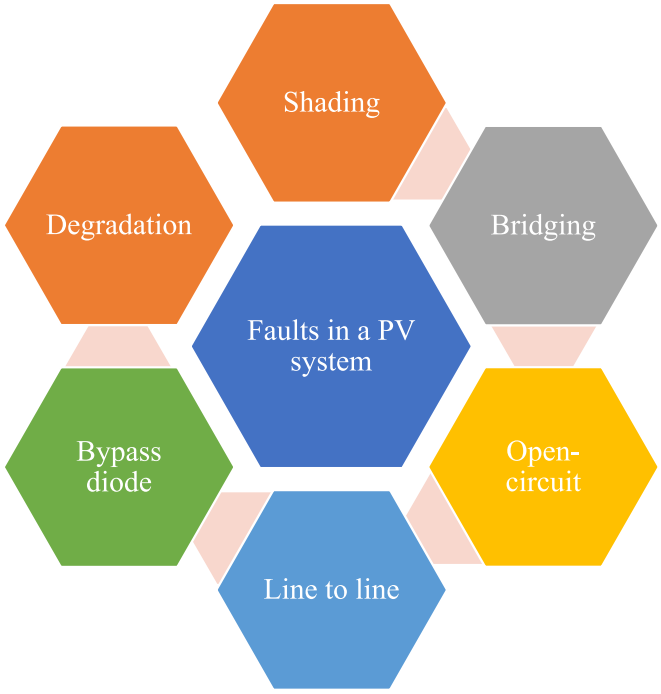


Fig. 5. Various faults common in a PV system.

there are also some minor ones, including weeds (H and B).

4.2.1. Fault detection

In this part, each type of fault is introduced, and information about a number of studies that have considered that type of fault is provided.

4.2.1.1. Shading. The performance of a PV system has a direct relationship with the incident solar radiation (Anderson et al., 2022). Therefore, when part or all of a module is shaded, its generated power goes down. Shading can originate from different things, including soil and dirt, objects such as trees, surrounding buildings, or even other panels.

One solution that can be employed to solve the shading problem is diode bypassing. Diode bypassing refers to isolating shaded modules from others. Another solution is controlling the modules' orientation to find the least shadow level and highest energy production, while finding the optimum arrangement (number of series and parallel modules) is another way to deal with the issue.

As an example of studies of shading detection in which a solution has been presented, Huang et al. (2019) provided an ANN with 1 hidden layer. Extreme machine learning was the training algorithm, in which the Bee Colony method was used. The resulting ANN was then used to detect shading by dust and dirt. Moreover, Adhya et al. (2022) took advantage of three newly presented machine learning algorithms, namely extreme gradient, categorical, and light gradient boosting approaches to diagnose a number of faults in a PV array system, including shadow. A Python 3.7 interface was used, and light gradient boosting was found to be the best approach for fault classification and detection. The method was more than 99.9% successful. Furthermore, Mellit and Kalogirou (2022) conducted similar research in which a variety of machine learning and deep learning approaches were evaluated for detecting and classifying faults in PV array systems. Multiple faults, including partial shading were considered. The investigation demonstrated that machine learning and deep learning were not significantly superior for fault classification, while ANN and advanced stacking approaches, respectively, were the most accurate method of each type. Either a physical model in which degradation rate is considered or machine learning methods were suggested for detecting faults.

Babasaki and Higuchi (Babasaki, Higuchi) used machine learning to categorize faults for a PV string. Three types of defects, shadowing from other sources, system breaking, and weeds, were considered. They found that using the proposed fault diagnosis approach would lower the operating and maintenance costs considerably. Another investigation in the field was carried out by Maaløe et al. (Maal ø e et al., 2020), where Bayes Theorem was used to cluster various types of faults originating from shading. The simulation procedure developed took advantage of I–V characteristics to find differences between various operating modes. Typically, degradation has been investigated in addition to shading faults in these studies.

4.2.1.2. Degradation. Like any other device, the performance of a PV system declines. This process, which is called degradation, comes from different issues in a solar module, such as breaking the glass, hot spots, bubbles, discoloration of the encapsulant, delamination, ribbon discoloration, etc. (Sangeetha et al., 2021). In general, it is estimated that, on average, the performance of a solar module reduces by around 0.8% every year (Kim et al., 2021). According to the information provided in (Kim et al., 2021), of the indicated items, hot spots make the highest contribution (33%) to the degradation. Not only do hot spots have a negative effect on the performance, but they also cause differences between the model prediction and reality, as discussed completely in the previous parts. Ribbon discoloration is considered to be the second significant factor, contributing 12%. Broken glass and encapsulant discoloration are in third and fourth places, respectively. They are responsible for 10% and 9% of the degradation, respectively.

Ali et al. (2020) conducted a study to investigate hot spot effects on PV module operation. In that study, photos taken by an infrared camera were employed, while several methods for parameters specification, as well as a variety of image processing techniques, including texture, RGB, OGH, and LBP were examined. Evaluation of different methods using a number of datasets showed that the support vector machine (SVM) made superior predictions to other machine learning approaches. Further, Raslan and Çam (Raslan, Ertu ğ rul) selected the decision tree approach to categorize defects in a PV system. Degradation, as well as dust, open and short circuits, was taken into account. The algorithm could identify the faults automatically. Another similar study was carried out by Dhimish (2021) using a more advanced method called discriminate classifiers that also gave more accurate predictions than conventional machine learning approaches like SVM. Winston et al. (2021) SVM and BPNN to find and categorize failures, such as degradation and hot spot faults. The method had five input parameters, i.e. PV temperature, short and open circuit values, impedance, solar radiation, and power losses. According to the results, SVM demonstrated a greater accuracy than BPNN.

4.2.1.3. Open circuit. If a wire in a PV system does not transfer the current, there is an open circuit. Open circuits arise from various issues such as fuses blowing, terminal problems, connection faults, and broken wires (Espinoza Trejo et al., 2018). As indicated in several references, such as (Daliento et al., 2017) an open-circuit is more harmful than a short-circuit because of the growth in the current. However, reviewing the literature has demonstrated that machine learning approaches have rarely been employed for open-circuit fault detection. In the few studied cases, it was not studied separately, and was investigated alongside other types of fault.

For instance, Alajmi et al. (Alajmi, Aljahdali, Alsaheel, Fattah, Alshehri) used a linear kernel based machine learning method to find faults arising from open circuits. In addition to open circuit, they also investigated short circuit and hot spot failures that were also detected by their proposed algorithm. Current-voltage curves were employed to develop the method. Badr et al. (2021) examined a number of widely-employed machine learning tools, including KNN and SVM, for which the Bayesian optimizer was used to achieve better classification

conditions. Open circuit was one of the faults investigated, while shading, MPPT, and line-to-line defects were other defects studied. Moreover, Dhibi et al. (2020) employed a number of machine learning approaches, including KNN, PCA, and Reduced-Kernel Random Forest (RKRF) to provide a means of detecting a number of faults in a PV system, such as open-circuit. Data collection was done by installing 9 sensors to capture current and voltage. The study also included shading and line to line faults in addition to open-circuit.

4.2.1.4. Line to line. When the ion content in the air reaches a threshold value, or when there is accidental contact between lines, line to line faults occur (Indira et al., 2021). This provides a path with low impedance through which current passes, and may cause serious harmful effects (Padilla et al., 2014). According to the explanations available in (Prasad et al., 2018), there are three methods of detecting of this fault, which are processing signals, machine learning, and combined approaches. Reviewing the literature has demonstrated that a greater number of studies have used machine learning techniques to detect this fault than others.

SVM was applied by Yi and Etemadi (Yi, Etemadi) to discriminate line to line faults in a PV string system. Different values of mismatch percentage were evaluated. A half supervised ladder network was used by Chen et al. (2021) to identify defects in a PV system. The network they developed was refined compared to conventional ones, providing better activation and loss functions, as well as optimization and convergence.

The investigation by Eskandari et al. (2020a) is an example of such studies, in which a number of different techniques for finding a suitable method of prediction, employing techniques such as SVM, KNN, and Naive Bayes. Eskandari et al. (Eskandari et al., 1134) optimized an SVM structure for robustness. The features were extracted from the current-voltage characteristics of the PV array, while the pattern search approach was employed to increase reliability. In addition, in another similar study, Eskandari et al. (2020b) evaluated other methods with higher accuracy of detection of this type of fault. Gao et al. (Gao and Wai, 2020) also used CNN to select faults from 10 defined modes, including the shading and line to line types. Their method was developed so that it can be trained in online.

4.2.1.5. Bridging. If there are some problems in the PV system structure or wiring, the resistance between two points might be low, leading to short circuits and damage to the PV system (Paul et al., 2016). This fault is usually detected by analyzing power voltage characteristic curves (Ul-Haq et al., 2020). The work done by Harrou et al. (2021) is an example of such investigations. In that reference, a variety of machine learning methods, such as Gaussian process regression, SVM, and kernel mapping method were examined to identify six types of fault, including bridging.

Moreover, Zaki et al. (2021) used CNN as a robust prediction tool to detect a number of defects in a PV system. The defects considered included bridging and line to line failures, while six cases were investigated to analyze the performance of the algorithm which was developed using the MATLAB interface. The results indicated the higher performance of the CNN found in comparison to conventional artificial intelligence tools, such as SVM.

As reviewing the literature has shown, CNN is considered to be one of the favorite and strongest means for detecting faults, including bridging gaps. In addition to reference (Zaki et al., 2021), CNN has also been used in a number of other studies, such as that by Waqar Akram et al. (2019). In reference (Akram et al., 2019), a number of faults, including open circuit, mismatch, bridging, and earth defects were found and classified using the CNN method. The method was found rapidly, employing EL images.

4.2.1.6. Bypass diode. As explained, when a number of modules are shaded, their performance declines considerably, affecting the performance of the whole PV system (Karuppusamy et al., 2021). By installing bypass diodes, this problem can be solved, and shaded modules or cells will be set aside (Mansoor et al., 2020). If these diodes develop a problem, the system performance is lowered (Joice et al., 2022).

Chine et al. (Costa et al., et al.) proposed an algorithm to detect a number of defects in a PV power generation unit. Several types of defect, such as bypasses, short circuits, open circuits, connections, and shadow faults were considered. Data for solar radiation and temperature was used, while a simulation approach was also employed. The chosen machine learning tool was artificial neural network. The proposed method could be regarded as a good way to detect PV failures on a large scale. Moreover, Li et al. (2021b)detected a number of faults such as bypass diode by using the performance characteristics curves. By conducting a thorough analysis, the machine learning tool with the highest performance was found from among six widely used methods. The methods examined included decision tree, KNN, SVM, ANN, and naive Bayesian classifiers. The evaluation covered various aspects, e.g., dimension reduction, distribution of classification tools. According to the analysis conducted, the best performance was seen for ANN. The reference (Harrou et al., 2021) is also another example of such research work.

4.2.1.7. Others. In addition to the major defects that have been investigated so far, there are also other types of failure in a PV system (Fig. 4), and machine learning methods have been employed to detect those faults as well. Most studies in this field have tried to find hot spots on the module, which is the sign of such types of failure.

The study by Hwang et al. (2021b) is an example of those investigations. In that study, a neural network of the convolutional type (CNN) was employed to detect and classify a number of failures in a PV system. The list of phenomena considered included various types of faults, including broken modules, cracks, and short circuits. In addition to CNN, photos obtained by an infrared imaging camera were also utilized.

Moreover, Kumar and Maheshwari (2021) took advantage of both unsupervised and supervised machine learning approaches to detect and classify a number of faults in a PV system. Characteristics curves in addition to electroluminescence imaging were employed to find a variety of defects, including interconnections and cracks. Zefri et al. (2022) applied the deep learning method in combination with unmanned aerial vehicles (UAV) to detect and categorize faults in a PV system, including bird droppings, dust, and open and short circuit faults. Five classes were defined for categorization.

Application of CNN to identify faults in a PV system was investigated by Cipriani et al. (2020), where the machine learning method was supported by non-destructive thermography. The method was indicated as fast and low cost. Shadows, bird droppings, and dust were among the failures that could be found and classified by the proposed algorithm.

Some studies have been also tried to detect faults related to the inverter. For instance, Fazai et al. (2019) used the integration of the generalized likelihood ratio test (GLRT) and Gaussian process regression (GPR) to detect and model a group of faults in a PV system, including inverter and partial shadow failures. The method worked by monitoring, current, voltage, and power, while the evaluation was done by considering the execution time, as well as false alarm and missed detection rates. Another example of using CNN to detect other faults in a PV unit, Malik et al. (Malik et al., 1390) took advantage of that to identify failures in the inverter of a grid-connected system. In the machine learning tool they developed, ReLU and AlexNet are included to give more robust performance. The acronym ReLU refers to the rectified linear unit function, while AlexNet was a CNN with 25 layers.

Identification of the failures arising from inverters in a PV system was also the topic of research by Gunda et al. (2020). In that study, using a large volume of corrective maintenance data gathered from PV sites, latent Dirichlet allocation (LDA) based structural topic modeling was

employed, while a variety of inverter faults were considered.

For a PV island, Manohar et al. (2019) used CNN to detect a number of faults, including unsymmetrical/symmetrical defects, as well as inverter problems. The algorithm could not only find the faults, but it could also classify them. Comparing the results with those of decision trees (DT) and SVM showed its superiority. In addition, short-circuit faults in a PV unit were detected using combination of CNN and compressing sensors in (Wang, Yang, Liu, Tu, Chen).

4.2.2. Fault diagnosis

Fault diagnosis refers to finding the origin of an observed fault (Momeni et al., 2019). The fault origin must be detected correctly to provide an appropriate remedy (Mansouri et al., 2021).

The study conducted by Wu and Sun (Yuchuan et al., 2581) is an example of the early studies in the field, where the forward error back propagation algorithm (BP) was employed to diagnose faults in a grid-connected PV system. Application of the proposed algorithm, which was based on signal processing, demonstrated its high potential. Rezgui et al. (Rezgui, Mouss, Mouss, Mouss, Benbouzid) provided a way to diagnose faults in a PV system. Their proposed approach utilized SVM, which was optimized using the k-nearest neighbor (KNN) method. In (Yuchuan et al., 2581), five types of fault, including device overheating and abnormal grid voltage readings were considered, while (Rezgui, Mouss, Mouss, Mouss, Benbouzid) covered short circuits, as well as bypassing and blocking diodes.

A logic block neural network assisted algorithm for diagnosing faults in a PV system connected to the grid was proposed by Mostefa Khelil et al. (Kara Mostefa Khelil et al., 2020). The algorithm dealt with two types of fault, which were string disconnections and short circuit failures. Their algorithm had four input variables. These were the highest voltage and current from the characteristics curve, in addition to the working temperature and solar radiation. In order to reduce the error and boost the generalization capability, Kapucu and Cubukcu (2021) introduced an algorithm in which ensemble machine learning was utilized. That was a combined algorithm to diagnose both short circuit and partial shading defects.

A neuro fuzzy model to diagnose faults in a photovoltaic unit was developed by Cabeza and Potts (2021), where both voltage and current defects were investigated. The application of a smart neural network of probabilistic type was analyzed in the research work of Basnet et al. (2020) to diagnose a number of faults in a PV power generation system. The investigation was performed during the winter season and PNN was employed as the machine learning approach. The four defects covered were line-to-line, arc, and short and open circuit conditions.

A framework to diagnose the failures in a PV unit was proposed by Hajji et al. (2021). That study used supervised machine learning in combination with principal component analysis of the system, which was of the grid-connected type. Two and three defects from the network and PV sides were taken into account, respectively. On the one hand, grid faults were inverter and connection failures. On the other hand, sensor, connection, and partial shading were the considered to be defects on the PV side. The MATLAB simulation interface was also employed by Abd el-Ghany et al. (Abd el-Ghany et al., 2021) to diagnose faults in a string of PV panels based on the trajectory of the current and voltage change rates. The method given could classify the failures for both complete and partial shading, in addition to array and cell connections.

Lin et al. (2022) proposed an algorithm for improved fault diagnosis which emphasized the compound faults of a PV array. That was an SE-ResNet network approach. The study considered a number of failures, including dust in addition to line-to-line, open circuit, abnormal degradation, and shadow on the modules. The effect of choosing the ANN type was investigated by Khelil et al. (Kara Mostefa Khelil et al., 2021), where RBF, PNN, generalized and back-propagation groups (GNN and BPNN) were considered. The best reaction was seen for RBF, while GRNN and BPNN were found to provide more accurate results by comparison with others. A grid-connected PV array was studied, and the list of considered faults included connection failures and short circuit current.

4.2.3. Solving the problems, reconfiguration, and management

In addition to the fault detection and diagnosis, machine learning and artificial intelligence tools can also be used to solve the problems and management of PV systems, and some studies have tried to use machine learning and artificial intelligence tools for this purpose.

Bendary et al. (2021) used the ANFIS method to detect, diagnose, and re-configure an array of PV panels. The algorithm was based on comparison of the measured values of voltage and current with the values given by ANFIS. The study covered overcurrent defects, which had been indicated as the main fault challenging the system. ANFIS was utilized by Solis-Cisneros et al. (2022) for optimal re-arrangement of PV panels, a part of which was in the shadow. The method was dynamic with no need to modify connections. Moreover, Zhang et al. (2021) applied reinforcement learning based on particle swarms to control and refine the performance of a PV plant when faults such as partial shading took place. The goal of the re-arrangement algorithm was to minimize the capacity and mileage costs, in addition to maximizing the power production of the PV farm.

In another investigation, Narayanaswamy et al. (Narayanaswamy, Ayyanar, Spanias, Tepedelenlioglu, Srinivasan) found a method to achieve the optimum topology for a PV string using neural networks. The method was used when there was partial shadow on the PV array, and aimed to give the maximum power production.

Combining a method based on the mathematics of chess, the Knight's tour method, Rezazadeh et al. (2022) provided a method for rearranging PV arrays in shadowed conditions. A variety of PV strings of different sizes were examined and the ability of the method to reach maximum power production was compared with other approaches, including the Skyscraper puzzle and SuDoKu techniques, which showed the efficiency of the Knight's tour method to deal effectively with partial shadowing defects. Another study was carried out by Rezazadeh et al. (2021) to overcome partial shadowing faults in a PV string. In that investigation, a methodology based on prime numbers was employed.

Further, the optimum re-configuration switching topology set was found by Iraji et al. (2018). The particle swarm optimization technique was employed, and determined the lowest number of switches such that the greatest ability to realize various desired topologies was met. Pappachan (Neelamkavil Pappachan, 2022) took advantage of the integrated Red Deer Moth Flame optimizer to re-configure PV strings. The method was found to be efficient, especially for large scale capacities where it was shown that conventional puzzle-based algorithms were not able to reach a totally reliable condition. The algorithm was defined so as to obtain the highest possible power.

In order to manage open and short circuit defects in a PV string system automatically, Murillo-Soto and Meza (2021) introduced an re-arrangement algorithm in the MATLAB Simulink interface. In addition to the above faults, degradation was also taken into account, while a huge power recovery was seen by employing a re-configuration strategy. Additionally, to overcome the negative impact of shadow on the performance of a PV array, the gray wolf optimization technique was utilized by Yousri et al. (2020). They demonstrated that the optimum condition for re-configuration of PV array could be achieved using this method. A comparison between the results obtained from that study and those using conventional approaches such as the enhanced SuDoKu and total cross-tied (TCT) methods demonstrated the superiority of machine learning and artificial tools over conventional techniques. This point was indicated not only in this study (Yousri et al., 2020), but has also been shown in other studies in the literature, including (Rezazadeh et al., 2022) and (Rezazadeh et al., 2021).

The various machine learning methods that have been used in studies are illustrated in Fig. 6. Moreover, a number of the most important error-related criteria that have been used to evaluate machine learning tools in

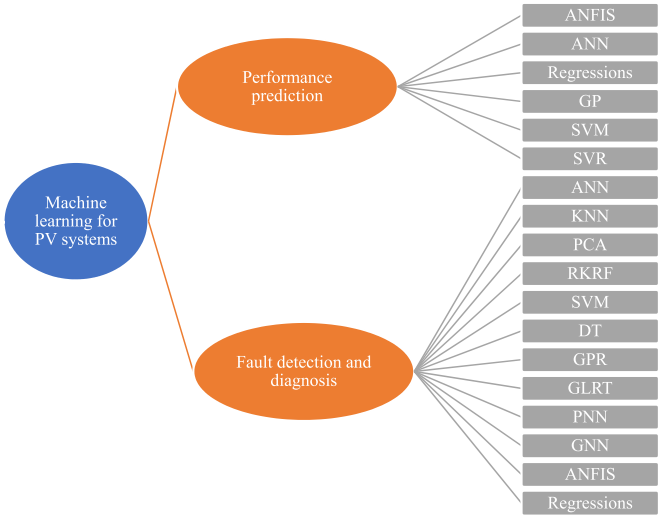


Fig. 6. The various machine learning methods that have been used in studies of performance prediction and fault detection in PV systems.

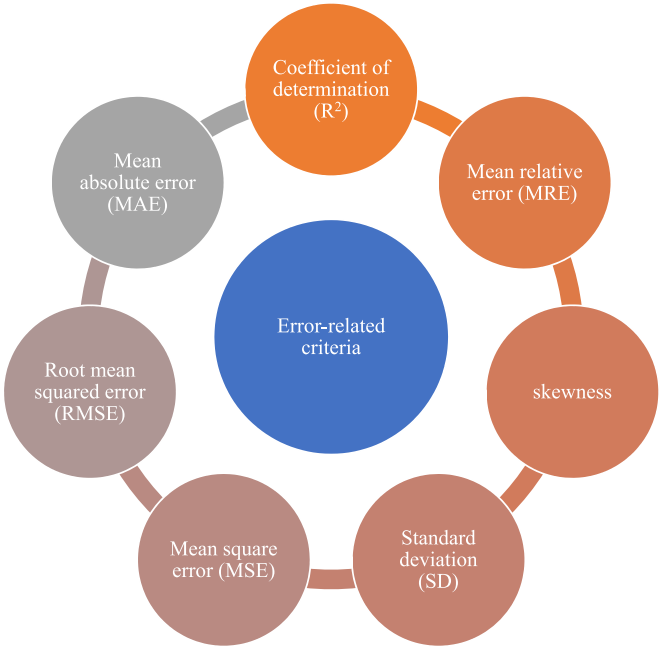


Fig. 7. Some of the most important error-related criteria that have been investigated to evaluate machine learning tools.

the investigations are listed in Fig. 7.

5. Role of machine learning approaches in achieving cleaner aspect and environmental targets

More accurate prediction of the power produced by a PV system leads to more precise sizing of the system. Moreover, when the working temperature of a module is determined more accurately, better thermal management could improve the system performance. As a result, a similar value of power could be supplied by a PV module with a smaller area and higher efficiency. The environmental impact of the PV during its life time is considerably reduced, while the goal of achieving more efficient energy systems is achieved. It should be noted that because a large part of the environmental emissions produced by a PV system arises from its production and disposal, major studies in the field of the environmental assessment of PV technologies have employed the life-cycle assessment (LCA) method for this purpose. LCA has been used in several studies, including those by Saedpanah et al. (2020), Fardi Asrami et al. (2021), Sumper et al. (2011), and Azzopardi and Mutale (2010). Moreover, there are several important reviews that have covered the application of machine learning approaches to PV systems, including Parida et al. (2011) and Hussien Rabaia et al. (2021).

In addition to the above research, the investigation by Sohani et al. (2022) is another example of research in which machine learning approaches were used to evaluate PV technologies from an environmental perspective. In that study, an ANN was developed to predict the photocurrent and thermal voltage of a 320 W poly crystalline solar module. Then, by taking advantage of the ANN models found, which were validated against experimental data, the impact of changing the photocurrent and thermal voltage of the diode on the CO_2 savings of the system was determined using a sensitivity analysis. The results demonstrated that by changing the photocurrent from 80% to 120% of the base value, the CO_2 savings grew by 32.6%. However, the same variation in the thermal voltage of the diode led to a 5.6% reduction in CO_2 savings. This implies that the CO_2 savings were more sensitive to the former than to the latter. In addition, the integration of concentrators and suitable cooling technologies for PV systems has been suggested as an effective way to improve energy and environmental performance.

Additionally, machine learning approaches could help to improve the operation of PV technologies, with less degradation. Moreover, the operational time of a PV system is increased by the application of suitable fault detection methods using machine learning methods. Consequently, the need to produce power in fossil-fuel based power plants and the number of modules requiring replacement during their life span both decline. Therefore, both life-cycle emission and electrical energy production have upward trends. Consequently, using machine learning methods could be very effective aids to realizing energy and environmental milestones.

6. Conclusions

This literature review has demonstrated that machine learning techniques are increasingly being adopted for both performance prediction and fault detection of PV systems. Neural network-based methods seem to be the dominant approach, complemented by other techniques such as SVMs.

A serious problem in the application of machine learning is that of overfitting. This refers to operation being acceptable within a limited range of inputs and unacceptable outside that range. To overcome the issue, the size of the input data could be increased, while the range of the effective parameters should be extended as far as possible. For cases beyond the scope of experiments, appropriate simulation methods are suggested.

Another critical point about using machine learning prediction tools is the connection between different subsystems, as well as the infrastructure deployed. Based on remarkable advances in the Internet of Things (IoT), sensors and systems could make a greater contribution to the field. In addition, using wireless networks and connections to robots responsible for cleaning the modules' surfaces is becoming more widespread. The cost imposed by infrastructure, including the computational cost and other devices, could be offset by enhancements in the energy and environmental benefits of the system. This would accelerate the cost reduction of future PV systems, as predicted in several references (Jahn et al., 2673). These strong positive impacts will lead to further applications of machine learning in PV solar farms, from large-scale units to small-scale PV installations, such as rooftop and BIPV/T systems.

CRediT authorship contribution statement

Ali Sohani: Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Hoseyn**

Sayyaadi: Conceptualization, Methodology, Investigation, Writing – review & editing, Supervision. **Cristina Cornaro:** Formal analysis, Writing – review & editing, Investigation, Methodology, Supervision. **Mohammad Hassan Shahverdian:** Data curation, Formal analysis, Investigation. **Marco Pierro:** Data curation, Formal analysis, Investigation, Visualization. **David Moser:** Data curation, Formal analysis, Investigation, Visualization. **Nader Karimi:** Data curation, Formal analysis, Investigation, Writing – review & editing. **Mohammad Hossein Doranehgard:** Investigation, Methodology, Writing – review & editing, Project administration. **Larry K.B. Li:** Formal analysis, Investigation, Methodology, Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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